Dr. Prajit T. RAJENDRAN

Curriculum Vitae

Highest qualification: Doctorate (Informatics), awarded 11/12/2025, Université Paris-Saclay, France
Google Scholar: scholar.google.com/citations?user=R2wEeSEAAAAJ | **LinkedIn:** linkedin.com/in/prajitrajendran
ORCID: 0000-0002-8283-9891 | **Web of Science ResearcherID:** PLS-3443-2026 | **GitHub:** https://github.com/Prajit-Rajendran
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RESEARCH PROFILE

Computer scientist and trustworthy-AI researcher with a PhD from Université Paris-Saclay (awarded 2025) and approximately six years of research experience across LLM safety, uncertainty quantification, active learning and evaluation. My doctoral research established methods that let a learning system decide, at run time and under a fixed labelling budget, which observations it does not understand and must escalate, combining stream-based active learning, unsupervised "unknown unknown" detection, and severity-aware sample prioritisation for safety-critical decisions. In parallel, a sustained collaboration in video coding and quality assessment has produced work on latency, energy and quality-constrained inference on resource-limited hardware. At CEA LIST, I develop white-box hallucination monitors for open-weight LLMs, multimodal out-of-distribution detection, adversarial multi-agent evaluations and reusable Python evaluation pipelines on HPC clusters. Published research on human oversight, severity-aware failure detection, safety shielding and federated active learning. Additional experience in efficient inference, distributed/federated learning concepts, and resource-constrained systems through cross-institutional collaborations.

Bibliometrics (indicative, as of 21/10/2026): 20 peer-reviewed conference/workshop papers, 1 journal article, ~250 citations (Google Scholar). All outputs are open access via arXiv, CEUR-WS, the HAL/CEA repository or gold open access.

EDUCATION

Ph.D. in Informatics (Computer Science) **01/11/2020 – 11/12/2025**
Université Paris-Saclay, France | Research conducted at CEA LIST, Palaiseau (Laboratoire d'Ingénierie des Systèmes / DILS).
Date of doctoral award: 11/12/2025.

- Thesis title: Expert-in-the-Loop Learning and Adaptation under Uncertain and Unpredictable Situations in AI-based Autonomous Systems
- Thesis topic: safety-aware and human-in-the-loop active learning for deep-learning-based autonomous systems including uncertainty estimation, unknown-unknown detection, and criticality-weighted data selection under streaming conditions.

M.Sc. in Information and Network Engineering **01/08/2018 – 30/11/2020**
KTH Royal Institute of Technology, Stockholm, Sweden.

- Specialisation in machine learning, signal processing and networked systems. Master's thesis carried out in industry at Scania CV AB.

B.Eng. in Electronics and Communication Engineering01/08/2014 – 30/06/2018
Ramaiah Institute of Technology, Visvesvaraya Technological University, Bengaluru, India.

PROFESSIONAL EXPERIENCE

Research Engineer — Trustworthy LLMs & AI Safety **01/04/2025 – present**
CEA LIST, Université Paris-Saclay, Palaiseau, France.

- Developed a white-box hallucination monitor based on attention-derived uncertainty signals (building on RAUQ), flagging potentially confidently wrong outputs of open-weight LLMs in one forward pass without repeated sampling or an external judge.
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- Designed adversarial multi-agent LLM scenarios with incentives for deception; building evaluations to measure whether monitoring and judging mechanisms detect misleading behavior.
- Built and maintained reusable Python pipelines for LLM-as-judge and factuality assessment, run as batch jobs on GPU clusters via Slurm, measuring accuracy, coverage and failure modes across prompts and model versions; analysed large interaction logs and reported mitigations to stakeholders.
- Worked with open-weight models and parameter-efficient fine-tuning workflows; developed research-to-engineering tooling using PyTorch, Hugging Face, LangChain and associated Python infrastructure.

Data Scientist **01/11/2023 – 31/05/2024**
Upfluence SAS, Lyon, France.

- Performed analysis of large-scale, high-throughput social data streams (Python, SQL) supporting commercial decision-making.
- Developed predictive models and analytical workflows exposing engagement, conversion and performance metrics to non-technical stakeholders.

Doctoral Researcher **01/11/2020 – 30/11/2023**

CEA LIST, Université Paris-Saclay, Palaiseau, France.

- Developed active-learning and uncertainty-aware systems that improve model performance iteratively under scarce labels, including stream-based active learning with adaptive uncertainty and diversity thresholds.
- Designed feedback-driven data-selection and evaluation strategies for classification and anomaly detection, incorporating perceptual ambiguity and criticality/severity weighting for safety-relevant decisions.
- Worked on federated learning in a stream-based active learning setting, maximising accuracy and label efficiency in a multi-client scenario.
- Developed unsupervised detection of "unknown unknowns" (inputs on which the model is confidently wrong) as a trigger for human intervention.
- Contributed to the EU Horizon 2020 TAILOR research network on trustworthy AI; published multiple peer-reviewed papers; chaired an IJCAI 2023 AISafety/SafeRL workshop session.

Research Intern — Connected Intelligence Team **01/01/2020 – 31/08/2020**

Scania CV AB, Stockholm, Sweden.

- Built vehicle-camera computer-vision/data-extraction pipelines, optimizing processing for near-real-time inference and limited onboard storage.
- Optimisation of data-processing workflows for near-real-time inference in edge and low-storage environments for practical deployment of ML on resource-constrained automotive hardware.

Software Engineering Intern — Engineering Development Group **01/01/2018 – 30/06/2018**

MathWorks Inc., Bengaluru, India.

- Re-architected core components of the Model Slicer verification tool (C++, MATLAB) for modularity and maintainability; translated user and engineering requirements into production-grade implementations.

PUBLICATIONS

Peer-reviewed journal articles

J1 V. V. Menon, **P. T. Rajendran**, C. Feldmann, K. Schoeffmann, M. Ghanbari, C. Timmerer. "JND-Aware Two-Pass Per-Title Encoding Scheme for Adaptive Live Streaming." IEEE Transactions on Circuits and Systems for Video Technology, 34(2):1281–1294, 2024. DOI: 10.1109/TCSVT.2023.3290725. Open access.
Significance: Establishes perceptually-driven, two-pass bitrate-ladder construction for live streaming; demonstrates that quality targets can be met at materially lower bitrate and energy cost, which is the applied basis of my work on inference under hard latency and resource budgets.

Peer-reviewed conference and workshop papers

C1 **P. T. Rajendran**, et al. "FedSALAT: Adaptive Buffer-Based Active Learning for Federated Data Streams." 2nd International Conference on Federated Learning and Intelligent Computing Systems (FLICS), 2026. Accepted / forthcoming.
Significance: Extends adaptive-threshold stream active learning to the federated setting, where each client must decide locally what is worth labelling.

C2 **P. T. Rajendran**, F. Arnez, H. Espinoza, A. Delaborde, C. Mraidha. "Oracle-Guided Soft Shielding for Safe Move Prediction in Chess." IEEE International Conference on Machine Learning and Applications (ICMLA), pp. 875–880, 2025. DOI: 10.1109/ICMLA66185.2025.00130. Preprint: arXiv:2603.08506.
Significance: Introduces a soft shielding layer that filters a learned policy's proposals against an oracle-derived safety signal without retraining the policy.

C3 **P. T. Rajendran**, H. Espinoza, A. Delaborde, C. Mraidha. "Stream-Based Active Learning with Adaptive Uncertainty and Diversity Thresholds." IEEE 33rd International Workshop on Machine Learning for Signal Processing (MLSP), pp. 1–6, 2023.
Significance: Core methodological contribution of the thesis: thresholds on uncertainty and diversity that adapt online, allowing a model to select what to learn from an unbounded stream under a fixed annotation budget.

C4 **P. T. Rajendran**, H. Espinoza, A. Delaborde, C. Mraidha. "Unsupervised Unknown Unknown Detection in Active Learning." AISafety/SafeRL Workshop, IJCAI 2023, CEUR-WS Vol-3505. Open access.
Significance: Detects inputs on which a model is confidently wrong, without supervised labels.

C5 **P. T. Rajendran**, G. Ollier, H. Espinoza, M. Adedjouma, A. Delaborde, C. Mraidha. "Safety-aware Active Learning with Perceptual Ambiguity and Criticality Assessment." AISafety Workshop, IJCAI 2022, CEUR-WS. Open access.
Significance: Weights sample selection by the severity of the consequence of error rather than by uncertainty alone, so that scarce annotation effort concentrates where failures matter most.

C6 **P. T. Rajendran**, H. Espinoza, A. Delaborde, C. Mraidha. "Human-in-the-Loop Learning Methods Toward Safe DL-Based Autonomous Systems: A Review." In Computer Safety, Reliability, and Security — SAFECOMP 2021 Workshops (WAISE), Lecture Notes in Computer Science vol. 12853, pp. 251–264, Springer, 2021. DOI: 10.1007/978-3-030-83906-2_20. Open access via HAL (cea.hal.science).
Significance: The most cited output of my record and a reference point in the human-in-the-loop safety literature; cited in subsequent surveys of human-in-the-loop ML for autonomous vehicles and in the Springer handbook literature on human controllability of AI systems.

C7 **P. T. Rajendran**, H. Espinoza, A. Delaborde, C. Mraidha. “Human-in-the-loop Learning for Safe Exploration through Anomaly Prediction and Intervention.” SafeAI Workshop, AAAI 2022, CEUR-WS. Open access.

C8 **P. T. Rajendran**, S. Afzal, V. V. Menon, C. Timmerer. “Energy-Quality-aware Variable Framerate Pareto-Front for Adaptive Video Streaming.” IEEE International Conference on Visual Communications and Image Processing (VCIP), pp. 1–5, 2024. DOI: 10.1109/VCIP63160.2024.10849829.
Significance: First-author contribution deriving the Pareto front between energy consumption and perceived quality, as a formal treatment of the compute/quality trade-off that governs whether adaptive inference is viable at the edge.

C9 A. Premkumar, **P. T. Rajendran**, V. V. Menon, C. Herglotz. “Content-Driven Frame-Level Bit Prediction for Rate Control in Versatile Video Coding.” IEEE International Symposium on Circuits and Systems (ISCAS), pp. 3911–3915, 2026. Preprint: arXiv:2602.06242.

C10 Y. Chen, B. Chen, H. Wei, A. C. Bovik, R. K. Mantiuk, A. Premkumar, **P. T. Rajendran**, V. V. Menon, et al. “ICME 2025 Generalizable HDR and SDR Video Quality Measurement Grand Challenge.” IEEE ICME Workshops (ICMEW), pp. 1–6, 2025. Preprint: arXiv:2506.22790.
Significance: Community benchmark report for a challenge in which my submitted method placed third; contributes a shared evaluation protocol for generalisable video quality measurement.

C11 A. Premkumar, **P. T. Rajendran**, V. V. Menon, A. Wieckowski, B. Bross, D. Marpe. “Quality-Aware Dynamic Resolution Adaptation Framework for Adaptive Video Streaming.” ACM Multimedia Systems Conference (MMSys), pp. 292–298, 2024.

C12 V. V. Menon, **P. T. Rajendran**, A. Premkumar, B. Bross, D. Marpe. “Video Super-Resolution for Optimized Bitrate and Green Online Streaming.” Picture Coding Symposium (PCS), pp. 1–5, 2024.

C13 V. V. Menon, **P. T. Rajendran**, R. Farahani, K. Schoeffmann, C. Timmerer. “Video Quality Assessment with Texture Information Fusion for Streaming Applications.” ACM Mile-High Video (MHV), pp. 1–6, 2024.

C14 V. V. Menon, A. Premkumar, **P. T. Rajendran**, A. Wieckowski, B. Bross, C. Timmerer, D. Marpe. “Energy-efficient Adaptive Video Streaming with Latency-Aware Dynamic Resolution Encoding.” ACM Mile-High Video (MHV), pp. 21–27, 2024.

C15 V. V. Menon, J. Zhu, **P. T. Rajendran**, S. Afzal, K. Schoeffmann, P. Le Callet, C. Timmerer. “Optimal Quality and Efficiency in Adaptive Live Streaming with JND-Aware Low-latency Encoding.” ACM Mile-High Video (MHV), pp. 61–67, 2024.

C16 V. V. Menon, A. Henkel, **P. T. Rajendran**, C. R. Helmrich, A. Wieckowski, B. Bross, C. Timmerer, D. Marpe. “All-Intra Rate Control Using Low Complexity Video Features for Versatile Video Coding.” IEEE International Conference on Image Processing (ICIP), pp. 2760–2764, 2023.

C17 V. V. Menon, J. Zhu, **P. T. Rajendran**, H. Amirpour, P. Le Callet, C. Timmerer. “Just Noticeable Difference-Aware Per-Scene Bitrate-Laddering for Adaptive Video Streaming.” IEEE International Conference on Multimedia and Expo (ICME), pp. 1673–1678, 2023.

C18 V. V. Menon, R. Farahani, **P. T. Rajendran**, M. Ghanbari, H. Hellwagner, C. Timmerer. “Transcoding Quality Prediction for Adaptive Video Streaming.” ACM Mile-High Video (MHV), pp. 103–109, 2023.

C19 H. Amirpour, **P. T. Rajendran**, V. V. Menon, M. Ghanbari, C. Timmerer. “Light-weight Video Encoding Complexity Prediction using Spatio-Temporal Features.” IEEE International Workshop on Multimedia Signal Processing (MMSP), pp. 1–6, 2022.

C20 V. V. Menon, H. Amirpour, **P. T. Rajendran**, M. Ghanbari, C. Timmerer. “Content-adaptive Encoder Preset Prediction for Adaptive Live Streaming.” Picture Coding Symposium (PCS), pp. 253–257, 2022.

Preprints and other open-access outputs

P1 A. Premkumar, **P. T. Rajendran**, V. V. Menon. “Learning Quality from Complexity and Structure: A Feature-Fused XGBoost Model for Video Quality Assessment.” arXiv:2506.09795, 2025.

P2 **P. T. Rajendran**, K. Creusy, V. Garnes. “Shorts on the Rise: Assessing the Effects of YouTube Shorts on Long-Form Video Content.” arXiv:2402.18208, 2024.

P3 V. V. Menon, S. Afzal, **P. T. Rajendran**, K. Schoeffmann, R. Prodan, C. Timmerer. “Content-Adaptive Variable Framerate Encoding Scheme for Green Live Streaming.” arXiv:2311.08074, 2023.

PRIZES AND AWARDS

- Third Prize, Generalizable HDR & SDR Video Quality Measurement Grand Challenge, IEEE ICME 2025 — international challenge sponsored by Amazon Prime Video; results published as C10 above.
- Runner-up, KTH Masters Challenge India, 2018 — national selection competition for admission and scholarship to KTH Royal Institute of Technology.

PEER-REVIEW AND ORGANISATION OF INTERNATIONAL CONFERENCES

- Peer reviewer, IEEE Transactions on Circuits and Systems for Video Technology (TCSVT) — 15 papers reviewed.
- Peer reviewer, International Conference on Multimodal Interaction (ICMI) 2026 — 5 papers reviewed.
- Peer reviewer, International Conference on Interactive Media Experiences (IMX) 2026 — 1 paper reviewed.
- Peer reviewer, Vietnam Journal of Computer Science 2026 — ongoing.

Verified reviewing record maintained under Web of Science ResearcherID PLS-3443-2026.

- Session Chair, AISafety and SafeRL Joint Workshop, IJCAI 2023 — chaired alongside presenting own accepted paper (Unsupervised Unknown Unknown Detection in Active Learning, C4 above).

PRESENTATIONS AT INTERNATIONAL CONFERENCES AND ADVANCED WORKSHOPS

- Oral presentation, IEEE FLICS 2026 — "FedSALAT: Adaptive Buffer-Based Active Learning for Federated Data Streams.", Valencia, Spain, June 2026.
- Poster presentation, IEEE ICMLA 2025 — "Oracle-Guided Soft Shielding for Safe Move Prediction in Chess.", Boca Raton, United States, December 2025.
- Poster presentation, IEEE MLSP 2023 — "Stream-Based Active Learning with Adaptive Uncertainty and Diversity Thresholds.", Rome, Italy, September 2023.
- Oral presentation, IJCAI 2023, AISafety and SafeRL Joint Workshop — "Unsupervised unknown unknown detection in active learning", Macau SAR, China, August 2023.
- Oral presentation, IJCAI-ECAI 2022, Workshop on Artificial Intelligence Safety (WAISE) — "Safety-aware Active Learning with Perceptual Ambiguity and Severity Assessment, Vienna, Austria, July 2022.
- Oral presentation, AAAI 2022, SafeAI workshop — "Human-in-the-loop Learning for Safe Exploration through Anomaly Prediction and Intervention", Virtual/Canada, February 2022.
- Oral Presentation, SAFECOMP 2021, Workshop on Artificial Intelligence Safety (WAISE) — "Human-in-the-loop learning methods toward safe DL-based autonomous systems: A review", Virtual/United Kingdom, September 2021.

SUPERVISING AND MENTORING ACTIVITIES

Teaching Assistant — Internetworking **01/09/2019 – 31/12/2019**

KTH Royal Institute of Technology, Stockholm, Sweden (paid position, M.Sc. programme in Information and Network Engineering).

- Ran and supervised weekly laboratory sessions for the Internetworking course, guiding students through practical networking exercises and troubleshooting their implementations.
- Graded lab reports and assignments, providing individual feedback to students on their technical work.

Informal research mentoring — Machine learning in video coding and streaming **2024 – present**

Sustained co-authorship with Amritha Premkumar (junior researcher) within the Klagenfurt/Fraunhofer HHI video-coding collaboration.

- Provided ongoing technical guidance across a five-paper collaboration (MMSys 2024, PCS 2024, VCIP 2024, arXiv:2506.09795, ISCAS 2026); authorship shifted from her supporting my first-authored work to her taking first authorship as her independence grew.

FUNDING RECEIVED

- PhD research was partially funded by TAILOR, an ICT-48 Network of AI Research Excellence Centers funded under the EU Horizon 2020 research and innovation programme under grant agreement No 952215.

ADVANCED TRAINING AND CONTINUING EDUCATION

- Introduction to Cooperative AI Programme — Cooperative AI Foundation. Game theoretic and multi-agent foundations of cooperation between AI systems and between AI systems and humans. [Q3 2026, Online]
- Technical AI Safety Programme — BlueDot Impact, 2024. Structured programme on AI alignment. [Q4 2024, Online]
- Oxford Machine Learning Summer School (OxML), 2023. [June 2023, Oxford, United Kingdom]
- Human-Aligned AI Summer School, 2022. [August 2022, Prague, Czech Republic]
- Eastern European Machine Learning Summer School (EEML), 2021. [July 2021, Budapest, Hungary]

CORE TECHNICAL SKILLS

- Data science: Python, SQL, Pandas, NumPy, PySpark, scikit-learn, XGBoost, statistical/analytical workflows, predictive modelling
- Machine learning: Active learning, uncertainty estimation, anomaly/OOD detection, classification, continual/stream learning, reinforcement learning, safety shielding
- NLP & LLMs: Hallucination/factuality evaluation, RAG, prompt engineering, intent classification, NER, question answering, automatic evaluation pipelines
- Engineering & data: PyTorch, TensorFlow, Hugging Face, LangChain, MLflow, FastAPI, REST, AWS, PostgreSQL, MongoDB, Git, Linux, HPC Slurm job scheduling on GPU clusters

LANGUAGES

- Languages: English (native), Malayalam (native), French (B1), Hindi, Kannada, Tamil, Bahasa Indonesia, Japanese (N3).

OTHER ITEMS OF INTEREST

- Mobility: education and research conducted across India, Sweden and France; established record of integrating into new national research systems.
- Interdisciplinary interest in cooperative AI and in European digital sovereignty in green, secure and trustworthy artificial intelligence.